

Ibrahim Al-Howaid

613-296-6065 | ialhowaid@gmail.com | al-howaid.me | github.com/IbraHIM-Howaid | Ottawa, ON | **Canadian Citizen**

PROJECTS

- Carleton University Simlab EMR** | *Python, Flask, PostgreSQL, ReportLab* | [Case Study](#) Mar. 2026 – Jun. 2026
- Built a **full-stack clinical charting EMR** in **Flask** and **PostgreSQL** for Carleton University's nursing simulation lab, where students charted on simulated patients across **24 assessment sections** (vitals, neuro, cardiovascular, MAR).
 - Secured the app with **role-based access control**, **CSRF protection**, and **session-based authentication** to separate student and instructor permissions.
 - Developed an instructor admin panel with **real-time student activity tracking** and online status, plus cohort and class lifecycle management with **archive locking**.
 - Engineered a **ReportLab PDF engine** that renders **2 to 3 page** patient assessments from live charting data and stamps every page footer with the charting student and the exporting instructor. Added **bulk export** that packages PDFs into a ZIP organized by student and patient.
 - Tailored features to course requirements with Carleton faculty and ran ongoing **testing and QA** under active use.
- Wireless Shop Doorbell** | *ESP32, ESP-NOW, C++, Arduino IDE, Fusion 360* | [GitHub](#) Feb. 2026 – Jun. 2026
- Wrote **C++ firmware** for two **ESP32** boards that communicate over **ESP-NOW** using direct MAC-addressed peer-to-peer packets, removing the need for a router or WiFi network.
 - Designed a **hybrid power scheme**: the button unit runs on a 3.36V LiFePO4 cell wired to the 3V3 pin to bypass the onboard regulator and sleeps in **deep sleep**, waking only on an **RTC-pin interrupt (GPIO33)** to send a packet. The always-listening receiver runs on wall power. The battery unit targets months of runtime.
 - Built the receiver's speaker driver with a **BC547 transistor** on GPIO25 (1k base resistor, 10k pull-down), moved from a Micro:bit prototype to ESP32 after hitting voltage limits, and collaborated on a **3D-printed weatherproof enclosure**.

HACKATHONS

- Hack the Hill III (uOttawa)**: Frontend developer on a team building an **AI course schedule maker** for uOttawa with natural-language preferences and RateMyProf summaries.
- Hack Club Lift-Off (Nokia)**: Built the **frontend** (UI/UX in HTML, CSS, and JavaScript) for Cosmic Brews, a cafe game where players decipher alien languages to serve customers.
- Hack Club Campfire (Kinaxis)**: Built a **game jam entry** about a mole mining underground while working in a team.

EDUCATION

- University of Ottawa** Sep. 2026 – Present
Honours Bachelor of Science in Computer Science (Co-op Stream) Ottawa, ON
- OCDSB SWIFT Program, SWIFT Launch Secondary** Feb. 2026 – Jun. 2026
Co-op Program, Kanata North Technology Park Ottawa, ON
- Selected for a **competitive co-op program** in Kanata North Technology Park focused on emerging technologies.
 - Received technical briefings from **BlackBerry QNX**, **Nokia**, **Telesat**, **Nordion**, **Ross Video**, and the **NRC Canadian Photonics Fabrication Centre** on embedded systems, 5G, and satellite communications.
- Sir Robert Borden High School** Sep. 2022 – Jun. 2026
High School Diploma Ottawa, ON

EXPERIENCE

- Youth in Policing Initiative** Dec. 2023 – Mar. 2024
Ottawa Police Service Ottawa, ON
- Handled confidential information in a police environment and completed workshops on law enforcement and community service.
- Teaching Assistant** Jun. 2022 – Sep. 2022
Terre-des-Jeunes School Ottawa, ON
- Managed course material distribution and answered inquiries from students, parents, and faculty for the Arabic-language Sunday school program.

CERTIFICATES

Emergency First Aid + Level C CPR + AED: St. John Ambulance | **AODA**: OSG, 2026 | **WHMIS**: Vector Solutions, 2026

TECHNICAL SKILLS

Languages: Python, JavaScript, C++, SQL, HTML/CSS
Frameworks, Libraries & Databases: Flask, React, Vite, Motion, ReportLab, PostgreSQL, SQLite
Embedded & Hardware: ESP32, ESP-NOW, deep-sleep power management, transistor driver circuits, Fusion 360
Tools: Git/GitHub, GitHub Actions, VS Code, Arduino IDE | **Spoken**: English, French, Arabic